Japan's Persistent Trade Deficit under a Weak Yen:

A Challenge to Traditional Exchange Rate Theory

ECON 521

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1. Introduction

Abenomics triggered a sharp yen depreciation after 2012, a trend reinforced by the Federal Reserve's tightening cycle after 2021. Under the Marshall–Lerner condition, such a depreciation—from 76 yen per dollar in 2011 to 157 in 2024—should improve the trade balance by raising export competitiveness and lowering import demand. However, Japan instead recorded persistent and historically large trade deficits, including –3.54 trillion yen in 2023. This outcome challenges the standard view that depreciation reliably strengthens trade performance.

BOJ data show that export prices in invoice currencies barely moved despite large exchange-rate swings, indicating strong pricing-to-market behavior and weak pass-through. At the same time, domestic prices rose: CPI increased from 92.7 to 112.5, and PPI fluctuated sharply between –4.6% and +10.6%, especially during the 2022 global cost surge. These trends imply that depreciation raised import and production costs more than it stimulated export volumes. Structural changes—such as extensive offshoring and Japan's reliance on dollar-denominated intermediate inputs—reduce the export elasticity necessary for the Marshall–Lerner condition to hold.

2. Research Question

RQ1. How has the USD/JPY exchange rate influenced Japan's monthly trade balance from 2010 to 2024?

RQ2. What additional factors—such as CPI, PPI, and rising import dependence—explain Japan's persistent trade deficits despite substantial yen depreciation?

3. Literature Reviews

First, decades of outward FDI have altered Japan's production network. Thorbecke (2022) and Sasaki & Yoshida (2017) show that offshoring shifted large portions of manufacturing abroad, increasing reliance on imported intermediate goods. Under this arrangement, depreciation raises the cost of imported components from overseas affiliates. Ono & Baak (2014) note that tighter integration with Asian supply chains, particularly China, further magnifies this sensitivity on the import side.

Second, Japanese exporters widely engage in pricing-to-market. SHIMIZU & Sato (2015) and Bahmani-Oskooee & Karamelikli (2018) find that firms keep export prices stable in invoice currencies, greatly limiting exchange-rate pass-through. As Kubo (2023) emphasizes, this behavior dampens competitiveness gains and weakens the J-curve response typically predicted after depreciation.

Combined, high import elasticity and low export elasticity mean that depreciation may worsen, rather than improve, the trade balance. Nishigaki (2025) argues that these structural constraints increasingly dominate Japan's external performance. Evidence from Neeraj & Trofimov (2025) suggests similar dynamics in services trade, contributing to persistent deficits even amid a significantly weaker yen.

4. Data and Variable Construction

Our analysis uses monthly Japanese macroeconomic and external sector data from 2010M1 to 2024M12, covering major structural episodes such as the Abenomics depreciation (2012–2015), the pandemic shock, and the post-COVID dollar surge (2021–2024). All variables are aligned at a monthly frequency and sourced from official institutions.

4.1 Variables and Sources

Net Exports (NX)

Defined as Exports − Imports (billion yen), obtained from the Ministry of Finance (MoF). NX serves as the dependent variable. Since raw values are extremely large, NX is rescaled into billion yen for readability.

Real Exchange Rate (RER)

The real exchange rate is constructed by adjusting the nominal USD/JPY rate for relative price levels between the United States and Japan:

$$RER_t = EXJPUS_t \times \frac{CPI_t^{US}}{CPI_t^{JPY}}$$

This measure reflects changes in Japan's price competitiveness by incorporating inflation differentials.

A higher RER indicates a real depreciation of the yen, making Japanese exports more competitive while increasing the domestic cost of imported goods..

CPI (Consumer Price Index)

From the Statistics Bureau of Japan (SBJ), indexed to 2020=100.

Captures demand-side inflation and potential nominal adjustments in import volumes.

PPI (Producer Price Index)

From the Bank of Japan (BOJ), indexed to 2020=100.

Measures imported-input cost pressures faced by Japanese manufacturers, a key factor behind Japan's structural trade deficits.

IPI (Industrial Production Index)

From the Ministry of Economy, Trade and Industry (METI).

Captures cyclical production activity and export capacity.

4.2 Descriptive Statistics

NX exhibits substantial volatility, with values ranging from –3.5 trillion yen to +1.1 trillion yen, and an average deficit of –348 billion yen. RER fluctuates from 185.6 to 452.5, reflecting large swings in yen valuation. CPI, PPI, and IPI show relatively modest variation but reveal persistent inflation and weak industrial output after 2020.

Variable	Mean	std	Min	Max	Obs
NX(billion yen)	-348.09	795.91	-3538.81	1109.73	180
RER	285.02	5.17	185.60	452.52	180
CPI	98.86	0.35	92.7	112.5	180
PPI	1.62	0.26	-4.6	10.6	180
IPI	108.70	0.39	87.6	116.9	180

The joint evolution of NX and EXJPUS shows a clear empirical pattern:

yen depreciation coincides with a worsening trade balance, consistent with the short-run import-cost channel. These preliminary patterns motivate the regression analysis in the next section.

5. Empirical Strategy

5.1 RQ1: How has the USD/JPY exchange rate affected Japan's monthly trade balance from 2010 to 2024?

$$NX_t = \alpha + \beta_1 REX_t + \beta_2 NX_{t-1} + \varepsilon_t$$

Variable	Coefficient	Std. Error	t-Statistic	Prob.
REX	-1.570520	0.775767	-2.024474	0.0444
NX(-1)	0.463934	0.067470	6.876130	0.0000
α	260.7232	221.4803	1.177185	0.2407

This regression examines how movements in the real USD/JPY exchange rate (REX) affect Japan's monthly net exports (NX) from 2010–2024, controlling for trade-flow persistence through a lagged NX term. The R-squared of 0.27 indicates that the exchange rate and past trade balance explain roughly one-quarter of the monthly variation, which is reasonable for high-frequency macro data.

Real exchange rate (REX)

The REX coefficient is –1.57 and significant at the 5% level (p = 0.044). A higher REX—yen depreciation—reduces the trade balance within the same month. This short-run deterioration arises because import prices adjust immediately, while export quantities respond slowly due to production and shipping lags. Japan's reliance on imported energy and intermediate goods amplifies this effect, reflecting the early phase of the J-curve.

Lagged trade balance NX(–1)

NX(–1) is positive and highly significant (0.464, $p < 0.001$), showing substantial persistence in monthly trade flows. This inertia stems from long-term contracts, production cycles, shipping schedules, and supply-chain rigidities.

Overall Interpretation

Overall, yen depreciation exerts a meaningful but short-run negative impact on Japan's trade balance, driven mainly by immediate import-cost pressures and slow export adjustment. Strong persistence in NX further indicates that trade imbalances adjust gradually. While the exchange rate is influential, it operates within a dynamic system shaped by pricing effects and structural inertia.

5.2 RQ2: What additional factors help explain Japan's persistent trade deficits despite substantial yen depreciation?

$$NX_t = \alpha + \beta_1 REX_t + \beta_2 CPI_t + \beta_3 PPI_t + \beta_4 IPI_t + \beta_5 NX_{t-1} + \varepsilon_t$$

Variable	Coefficient	Std. Error	t-Statistic	Prob.
REX	-6.800153	2.502955	-2.716849	0.0073
CPI	110.0499	35.72051	3.080859	0.0024
PPI	-74.22730	16.84150	-4.407405	0.0000
IPI	24.60652	10.60757	2.319714	0.0215
NX(-1)	0.255732	0.072968	3.504721	0.0006
α	-11754.23	3346.444	-3.512455	0.0006

This extended OLS model adds domestic price indicators and industrial production to explain why trade deficits persist even when the yen weakens. The adjusted R-squared of 0.366 indicates a notable improvement in explanatory power.

Real Exchange Rate (REX)

The coefficient (–6.80, $p = 0.0073$) confirms that yen depreciation worsens the trade balance in the short run. Higher import costs—especially for energy and raw materials—outweigh potential export gains.

Consumer Price Index (CPI)

CPI positively affects NX (≈110, $p = 0.002$). Higher consumer prices may suppress real import demand or coincide with stronger nominal export revenues during depreciation episodes.

Producer Price Index (PPI)

PPI has a large negative impact (–74.2, $p < 0.001$). Rising producer costs reduce export competitiveness and increase dependence on imported inputs, a key reason Japan's deficits persist.

Industrial Production Index (IPI)

IPI is positive and significant (24.6, $p = 0.0215$). Stronger production boosts export volumes and reduces the need for imported components.

Lagged NX(–1)

The lagged term remains significant (0.256), again indicating persistence in trade flows.

Overall Interpretation

Yen depreciation alone does not reverse Japan's trade deficits. Structural factors—such as rising production costs, imported-input dependence, and fluctuations in industrial output—dominate short-run exchange rate effects. Persistent inertia in NX further slows adjustment. These combined forces offset the competitive benefits of a weaker yen and help explain Japan's ongoing deficits.

7.Conclusion

This study examines how movements in the real USD/JPY exchange rate and domestic economic conditions affect Japan's monthly trade balance from 2010 to 2024. Across both models, the real exchange rate has a clear short-run effect, while structural domestic factors exert stronger and more persistent influence.

The first specification shows that real yen depreciation worsens the trade balance within the same month, driven by Japan's dependence on imported energy and intermediate goods. Import costs rise immediately, whereas export volumes adjust slowly due to production and shipping lags. The significant lagged NX term highlights strong persistence in trade flows and the gradual nature of adjustment.

The expanded model strengthens this interpretation. Even after controlling for exchange rate movements, producer-side inflation and industrial output explain most of the monthly variation in NX. Higher producer prices sharply weaken the trade balance by raising firms' cost burdens, while stronger industrial production improves NX by boosting export capacity. Consumer prices have a smaller positive association, likely through lower real import demand. Overall, the results show that yen depreciation cannot offset Japan's trade deficits when domestic cost pressures and production constraints remain binding.

Overall Implications

The empirical evidence reveals a structural foundation for Japan's persistent trade deficits. Although exchange-rate movements matter, they operate within a system where dependence on imported inputs, high producer costs, and weak industrial output largely determine short-run trade performance. With energy and intermediate goods still heavily import-reliant, real depreciation raises production costs more quickly than it boosts export volumes.

These results imply that policies centered solely on weakening the yen are unlikely to deliver lasting trade surpluses. Meaningful improvement requires strengthening domestic industrial capacity, reducing reliance on imported inputs, and easing producer-side cost pressures. Without addressing these structural constraints, even substantial yen depreciation will have limited ability to correct Japan's trade imbalance.

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